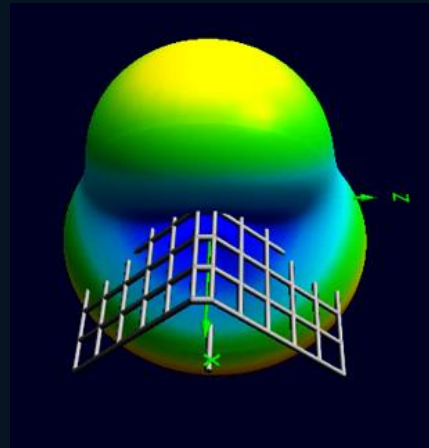
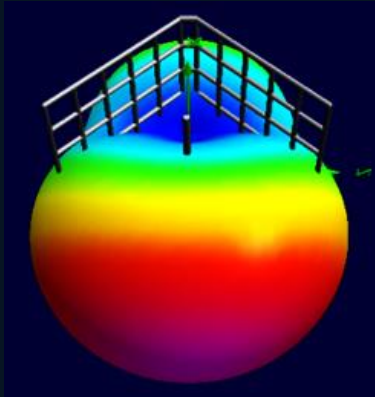
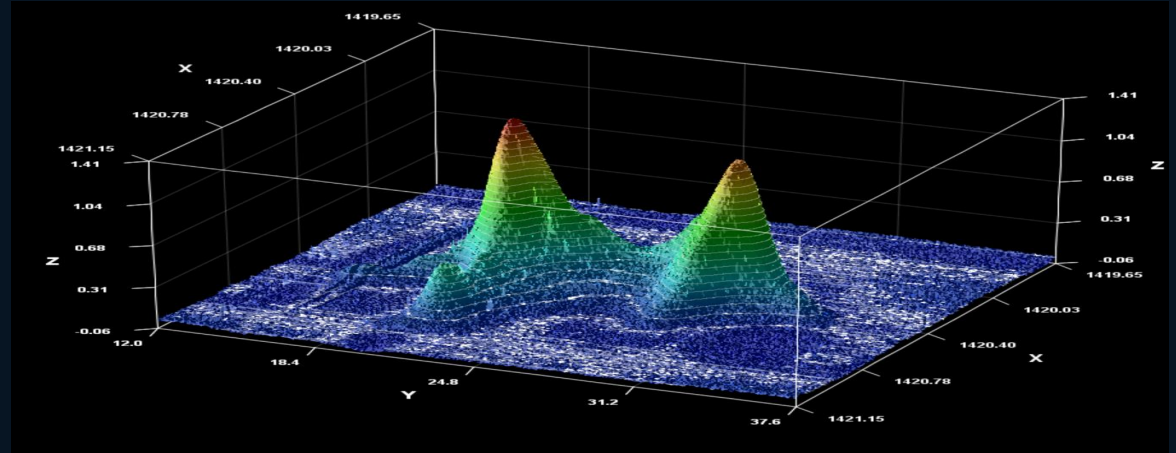
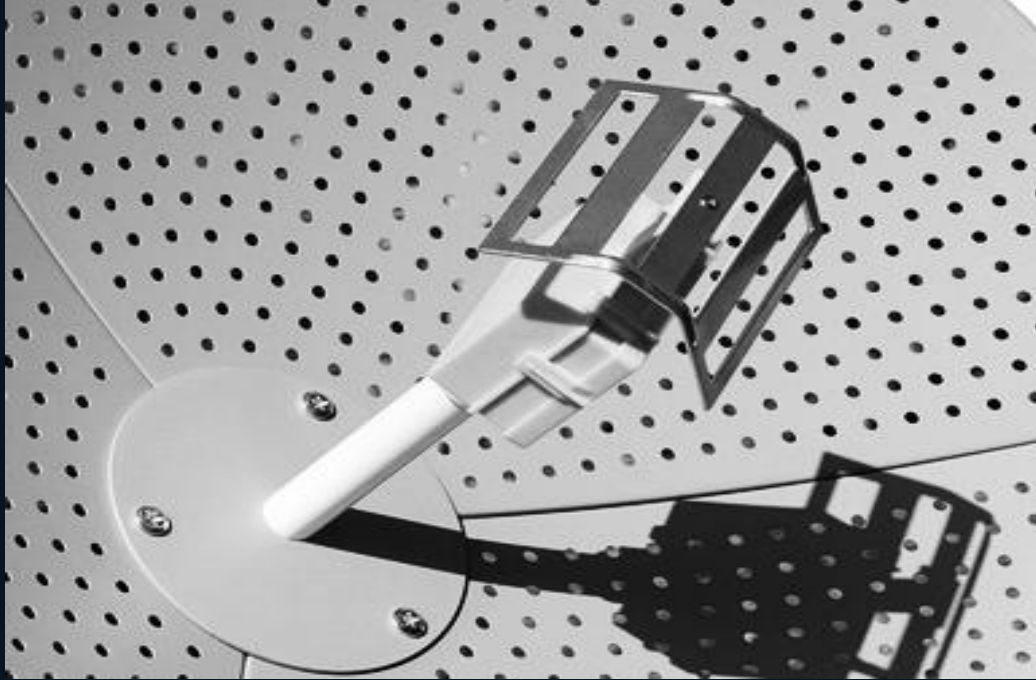
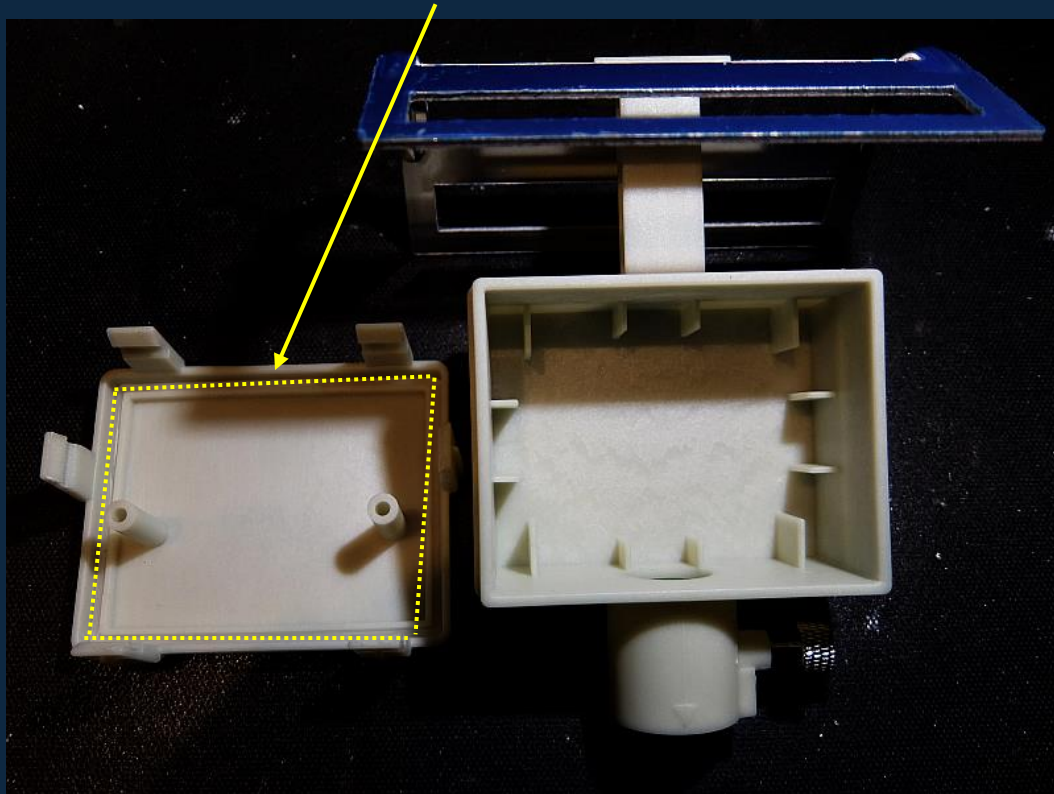


Kraken Discovery Dish 1420 MHz Feed Evaluation using a 1m WiFi Parabolic Reflector

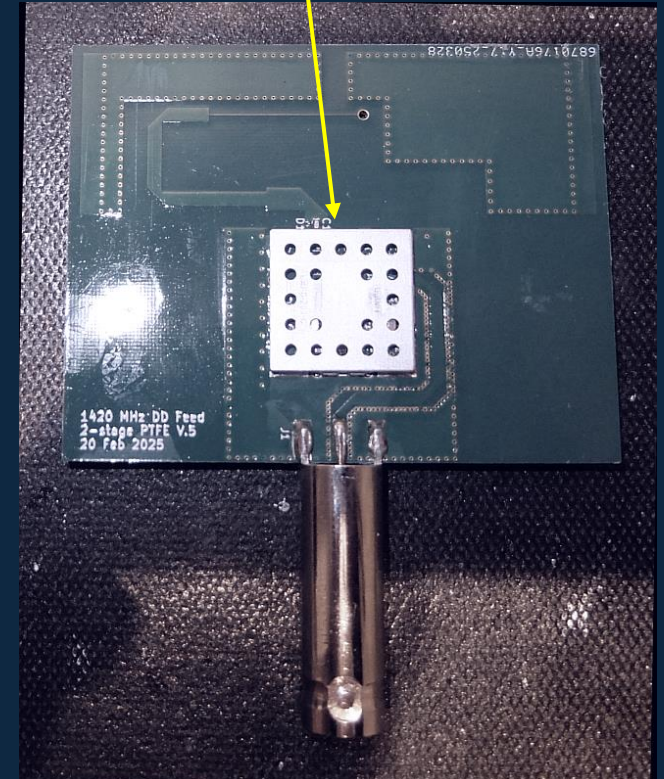
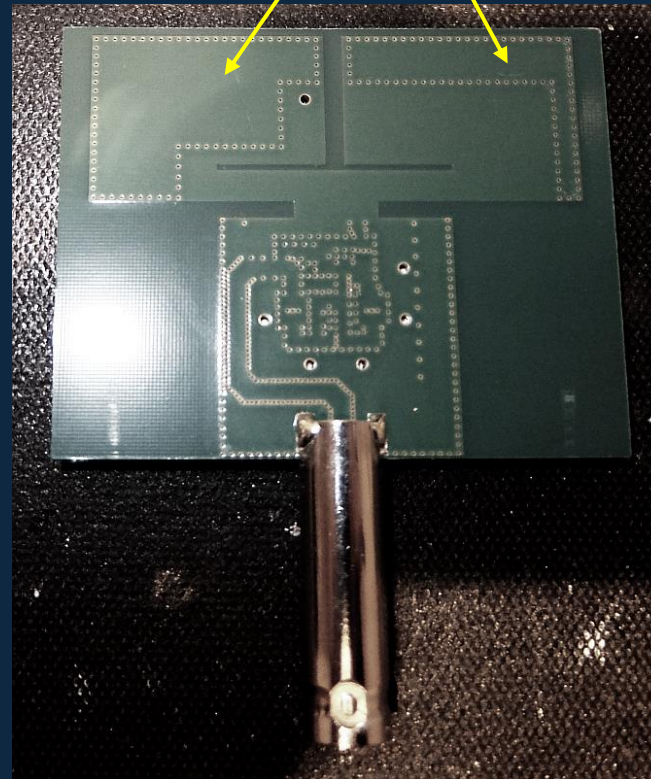


Discovery Dish 1420 MHz Feed Evaluation Construction

Box has weather-seal around perimeter



Antenna is closely coupled to LNA
to Minimize (loss) Noise Figure



Kraken Discovery Dish 1420 MHz Feed Evaluation LNA

Qorvo QPL9547 0.1GHz to 6.0GHz Ultra Low-Noise LNA

Qorvo QPL9547 0.1GHz to 6.0GHz Ultra-Low-Noise LNA is a high-linearity, ultra-low noise amplifier optimized for 5G Massive MIMO (Multiple Input Multiple Output) applications. At 1.9GHz, the amplifier typically provides 19.5dB gain, +39dBm OIP3 at a 65mA bias setting, and a noise figure of 0.3dB.

Key Features

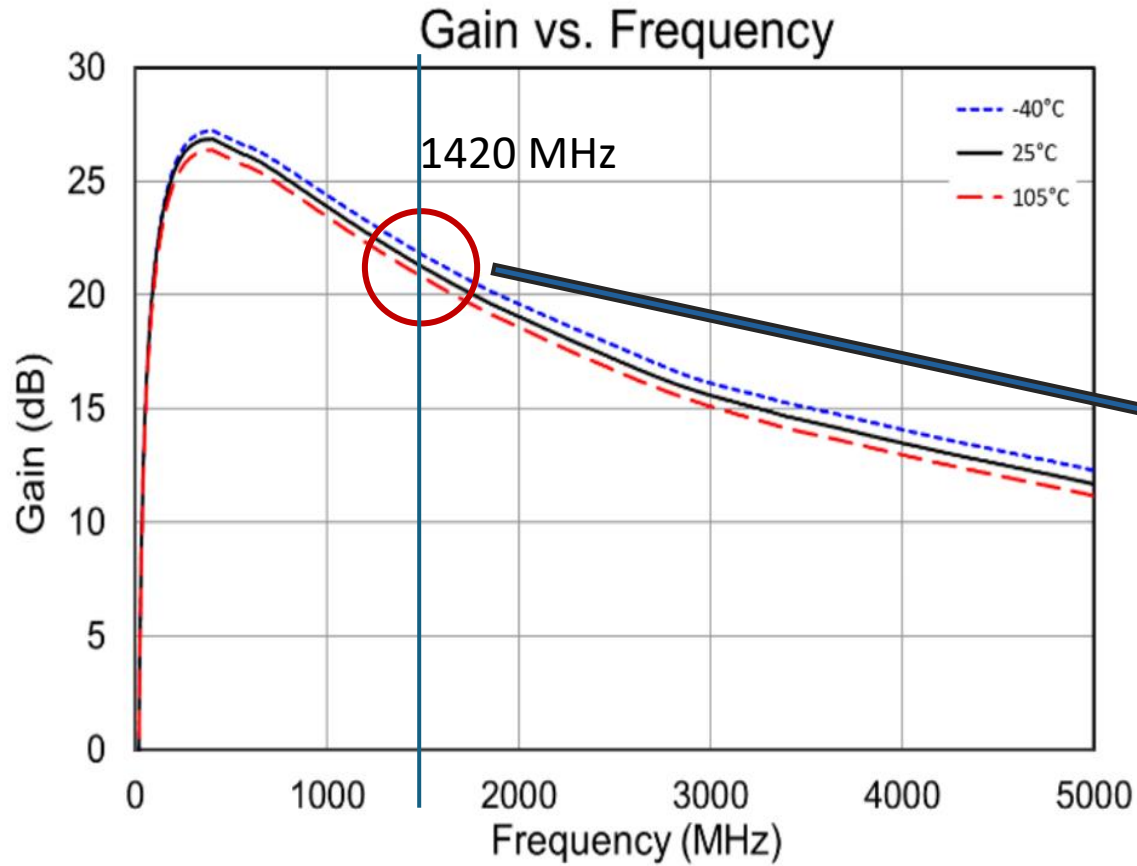
- 0.1-6 GHz Operational Bandwidth
- **Wide bandwidth: operates from 0.1GHz to 6 GHz**
- **Ultra-low noise: 0.3 db NF @ 1.9 GHz**
- **Gain: 19.5 dB typ.**
- **OIP3: +39 dBm typ.**

Qorvo QPL9547 0.1GHz to 6.0GHz Ultra Low-Noise LNA



Kraken Discovery Dish 1420 MHz Feed Evaluation LNA

Qorvo QPL 9547 LNA Information Gain vs Temperature Stability

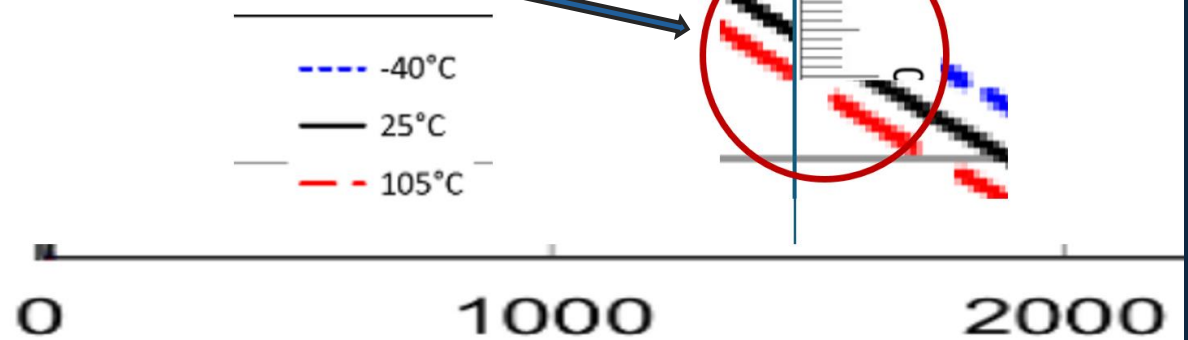


Gain vs. Frequency

QORVO®

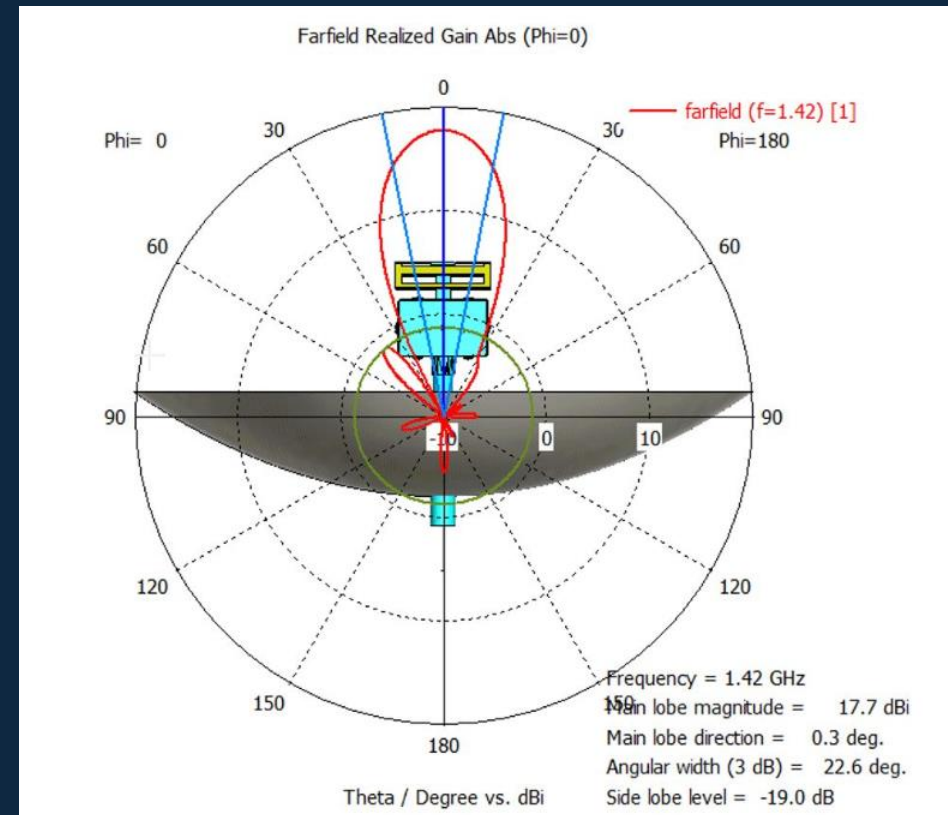
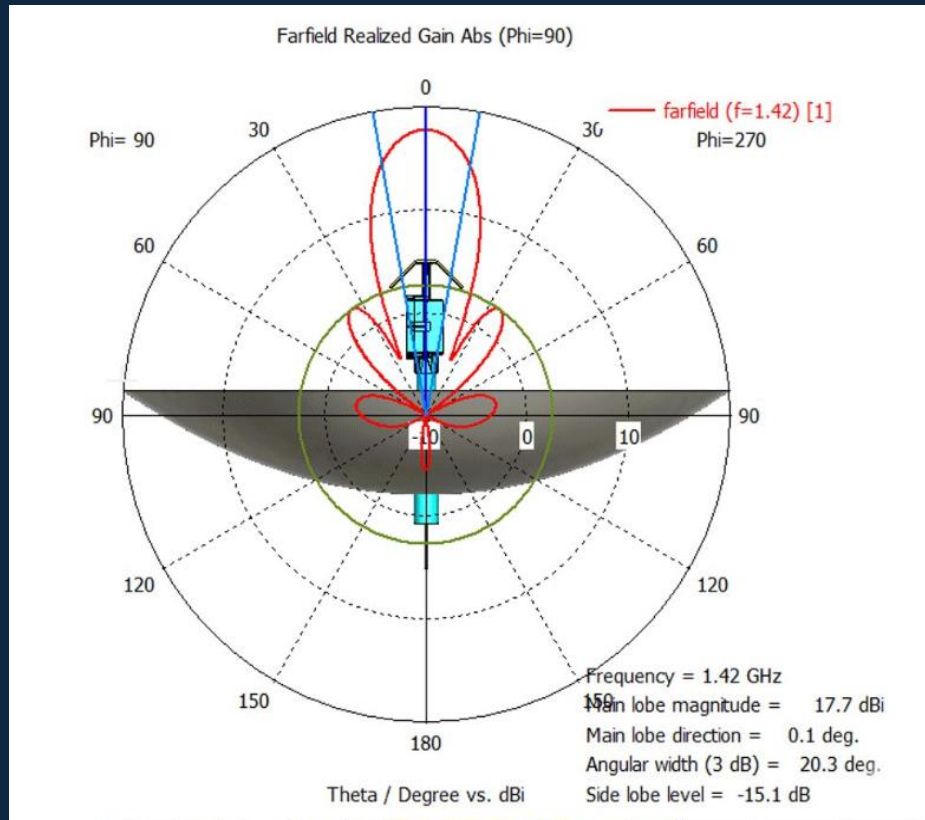
QPL9547
Ultra Low-Noise Amplifier

Gain vs Temperature
1.10 dB / 145° C = -0.075 dB / +10° C
0.45 dB / 80° C = -0.056 dB / +10° C



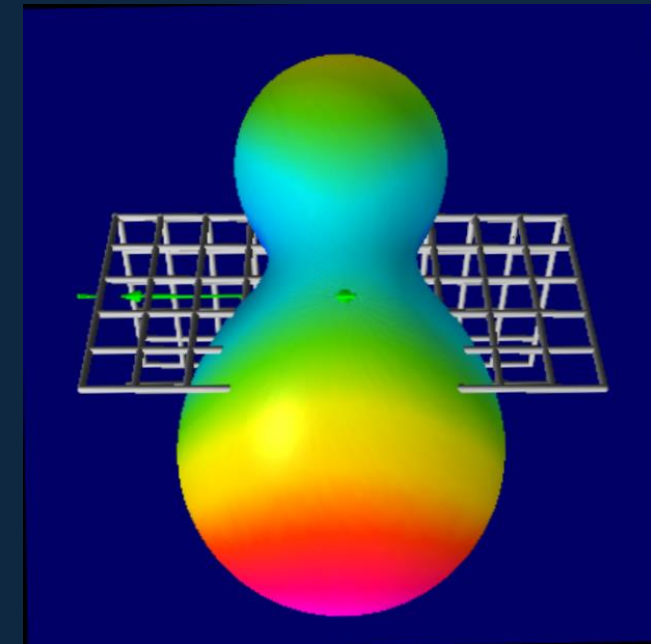
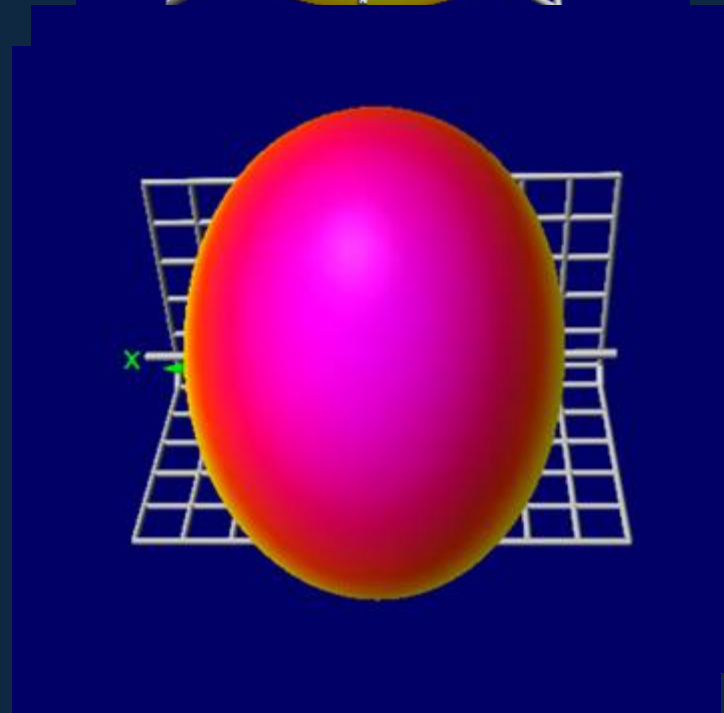
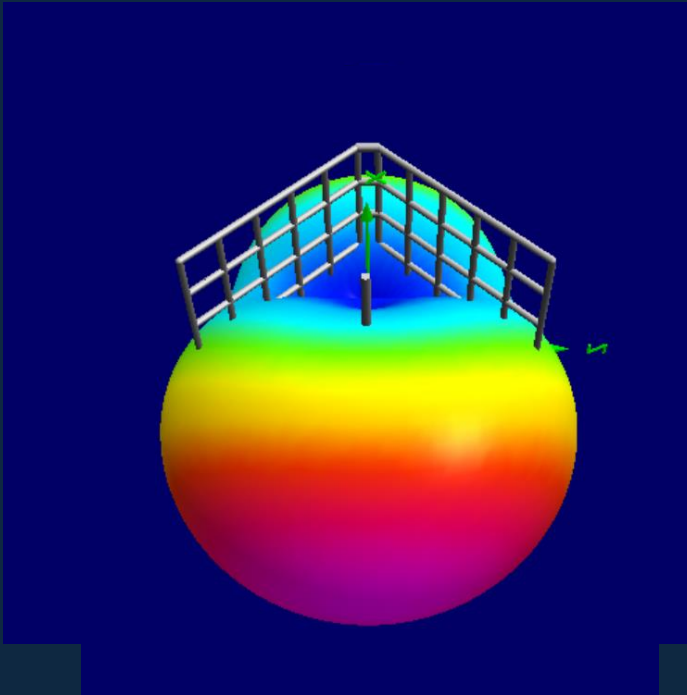
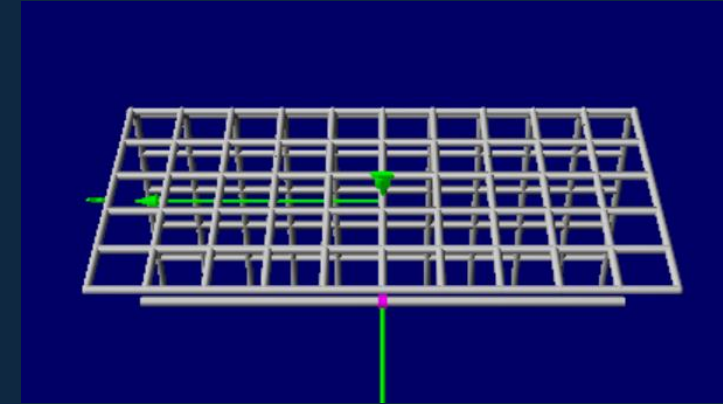
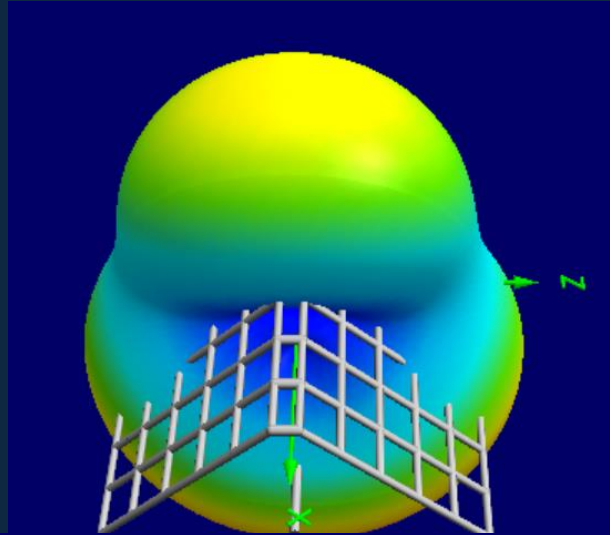
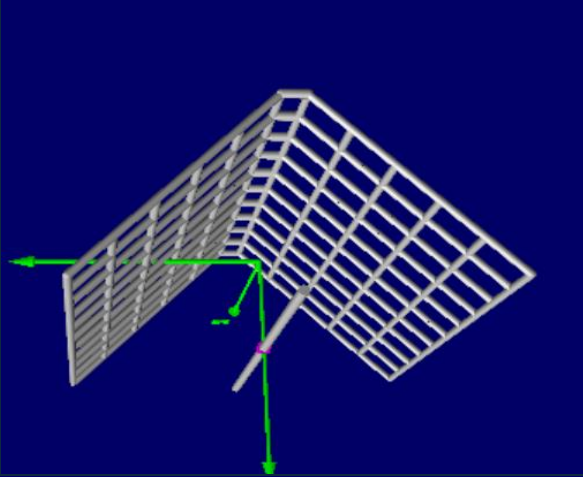
Kraken Discovery Dish 1420 MHz Feed Evaluation

Kraken information : Antenna Beam Patterns



Kraken Discovery Dish 1420 MHz Feed Evaluation
4NEC2 Numerical Electromagnetics Code Dipole/Reflector Analysis
The actual antenna is more complex than this simple dipole .

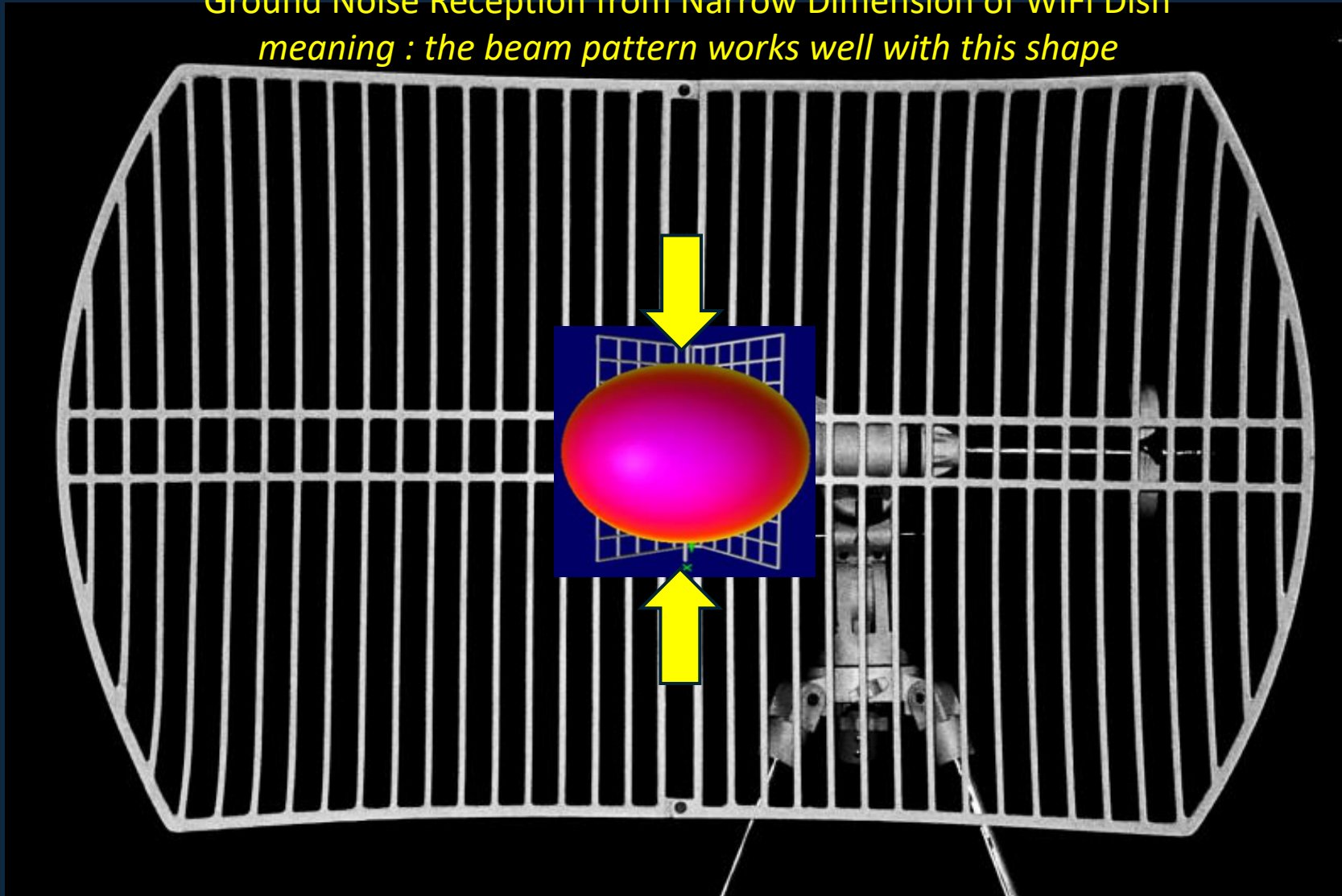
alex petit



Kraken Discovery Dish 1420 MHz Feed Evaluation
4NEC2 Numerical Electromagnetics Code Dipole/Reflector Analysis
The actual antenna is more complex than this simple dipole .

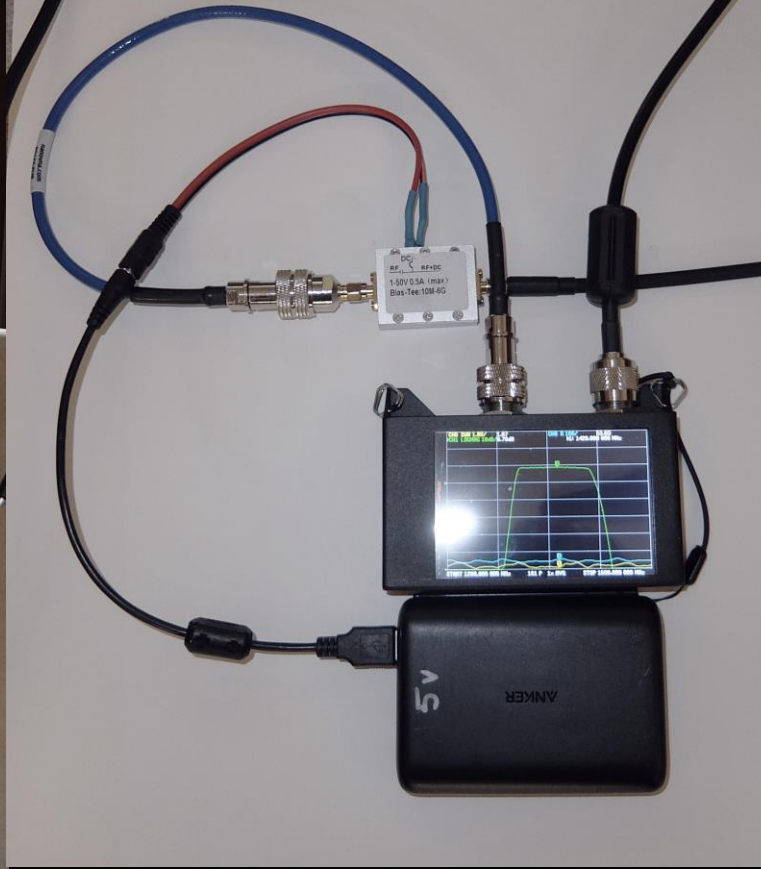
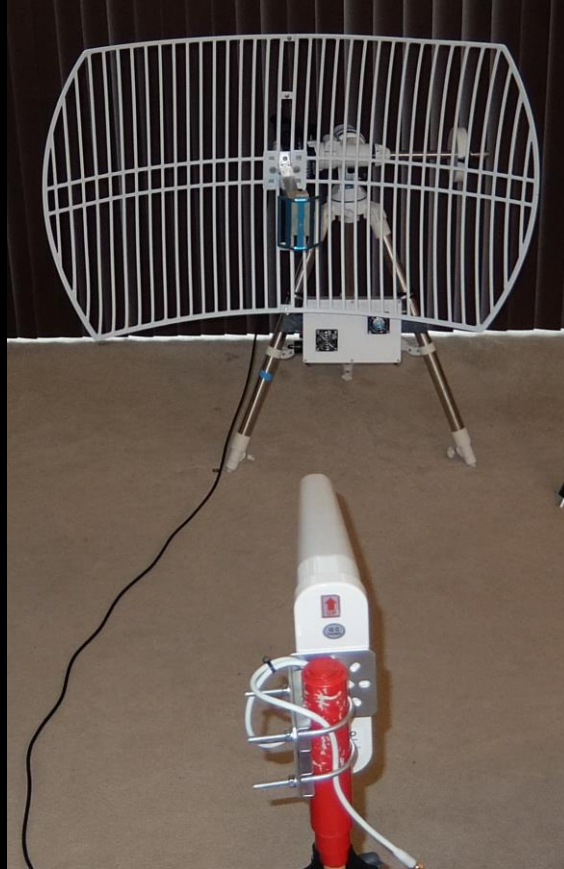
alex petit

Narrow Out-of-Axis Beam Pattern Minimizes
Ground Noise Reception from Narrow Dimension of WiFi Dish
meaning : the beam pattern works well with this shape



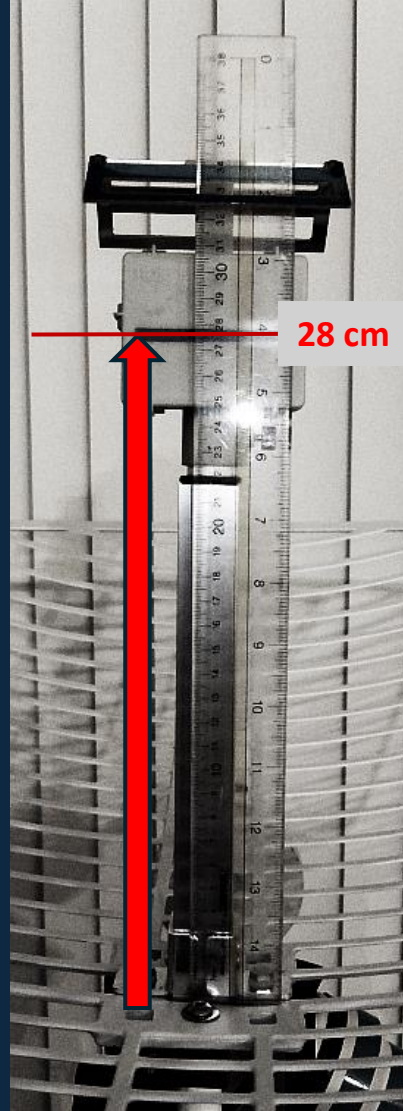
Kraken Discovery Dish 1420 MHz Feed Evaluation

Band Pass Filter Characteristics : 1380 – 1490 MHz

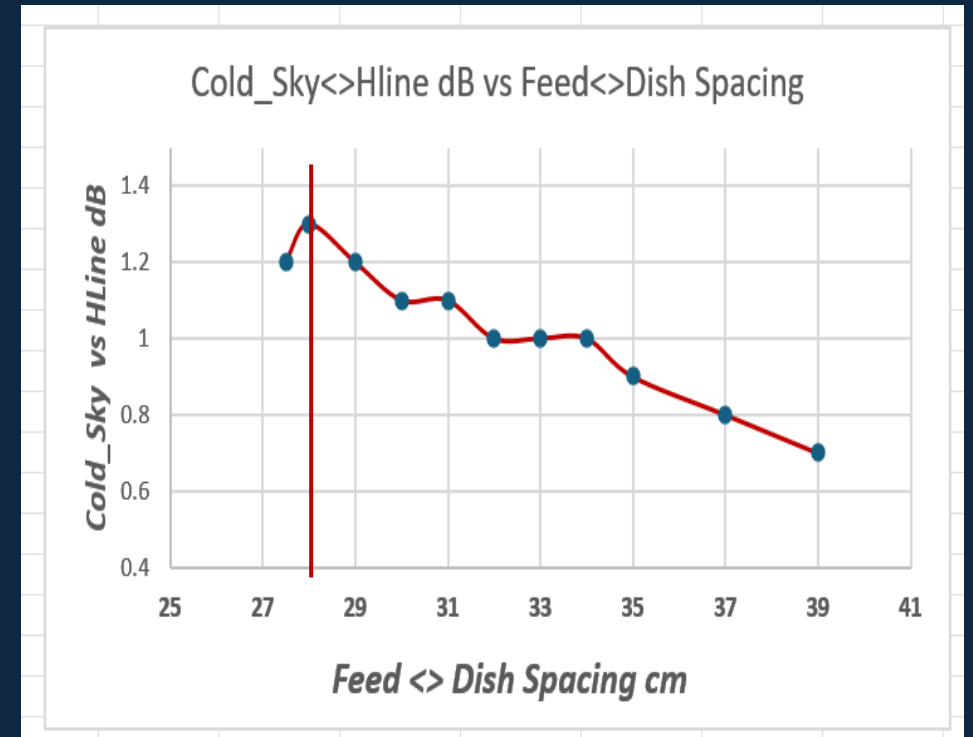


Kraken Discovery Dish 1420 MHz Feed Evaluation

WiFi Dish<>Feed Spacing Optimization

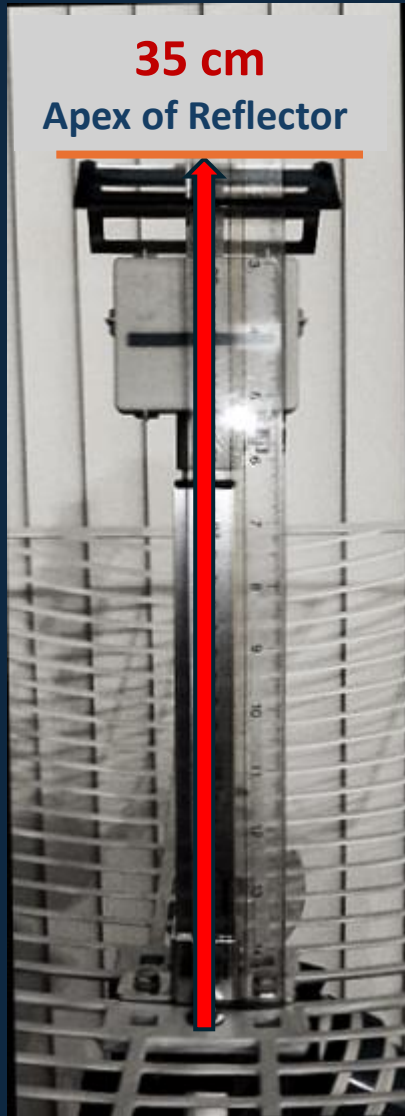


Nooelec 1m WiFi Grid Dish
Discovery Dish 1420 MHz Dipole Feed
Discovery Dish H1 internal LNA



Kraken Discovery Dish 1420 MHz Feed Evaluation

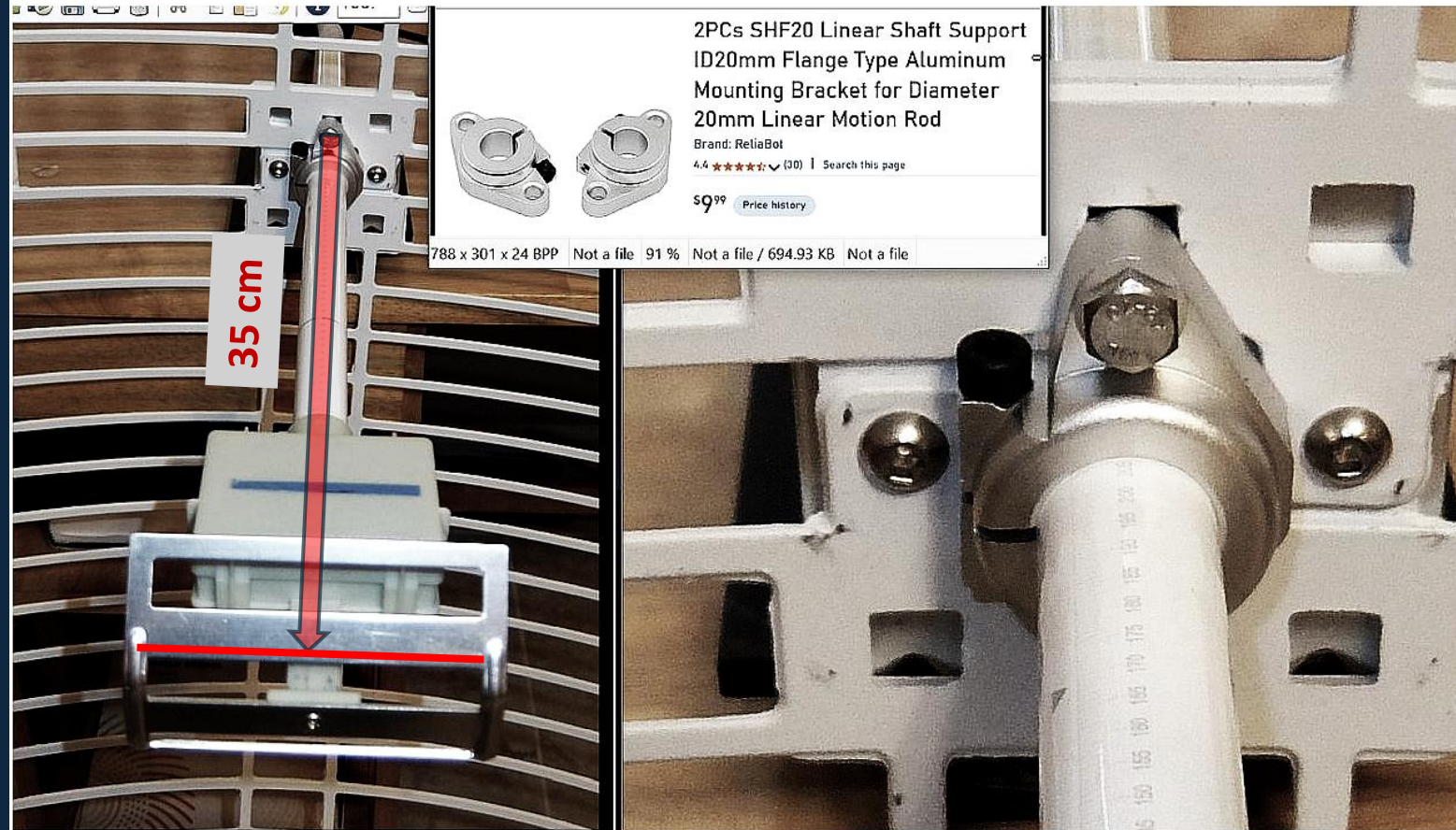
WiFi Dish<>Feed Attachment Flange



SHF20 Linear Shaft Support
ID 20mm Flange Type
Aluminum Mounting Bracket for
20mm Diameter Linear Motion Rod



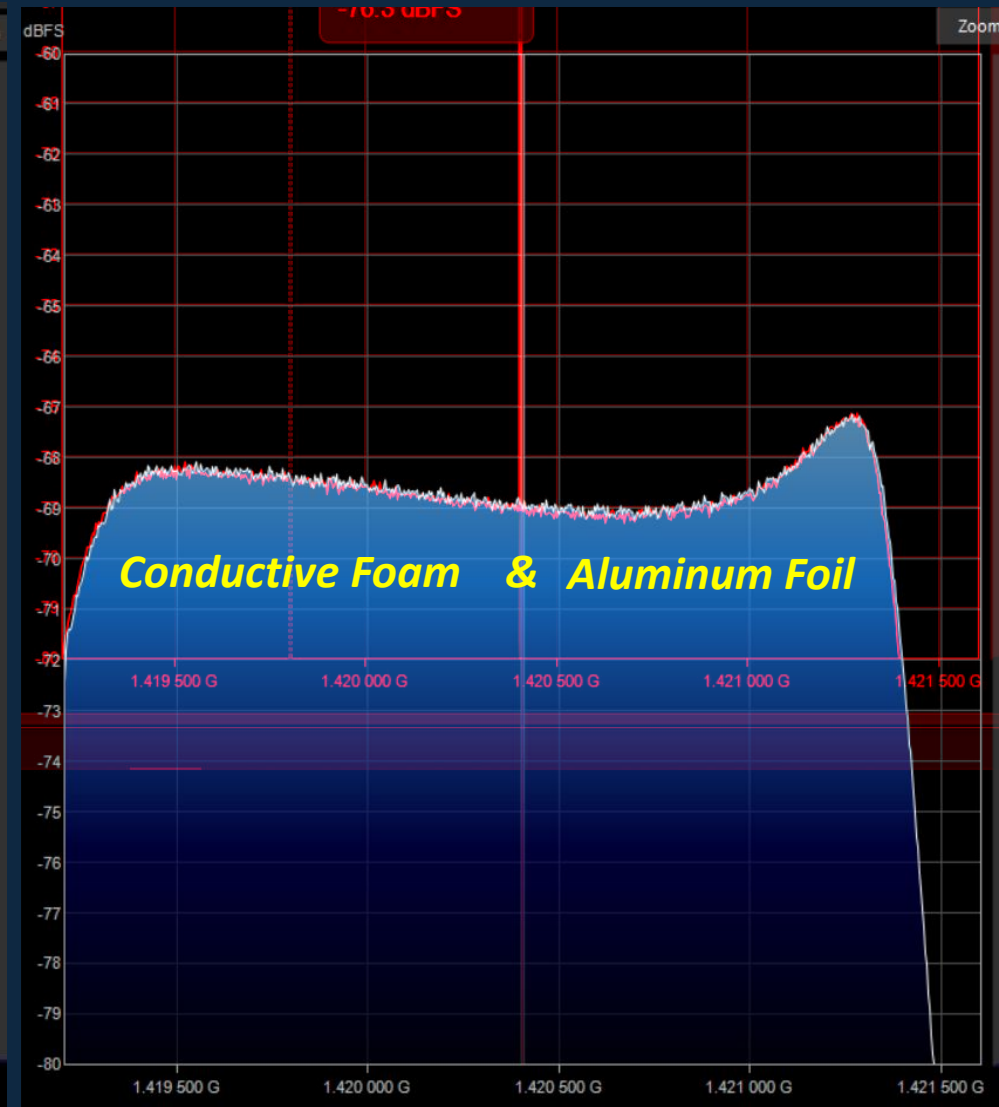
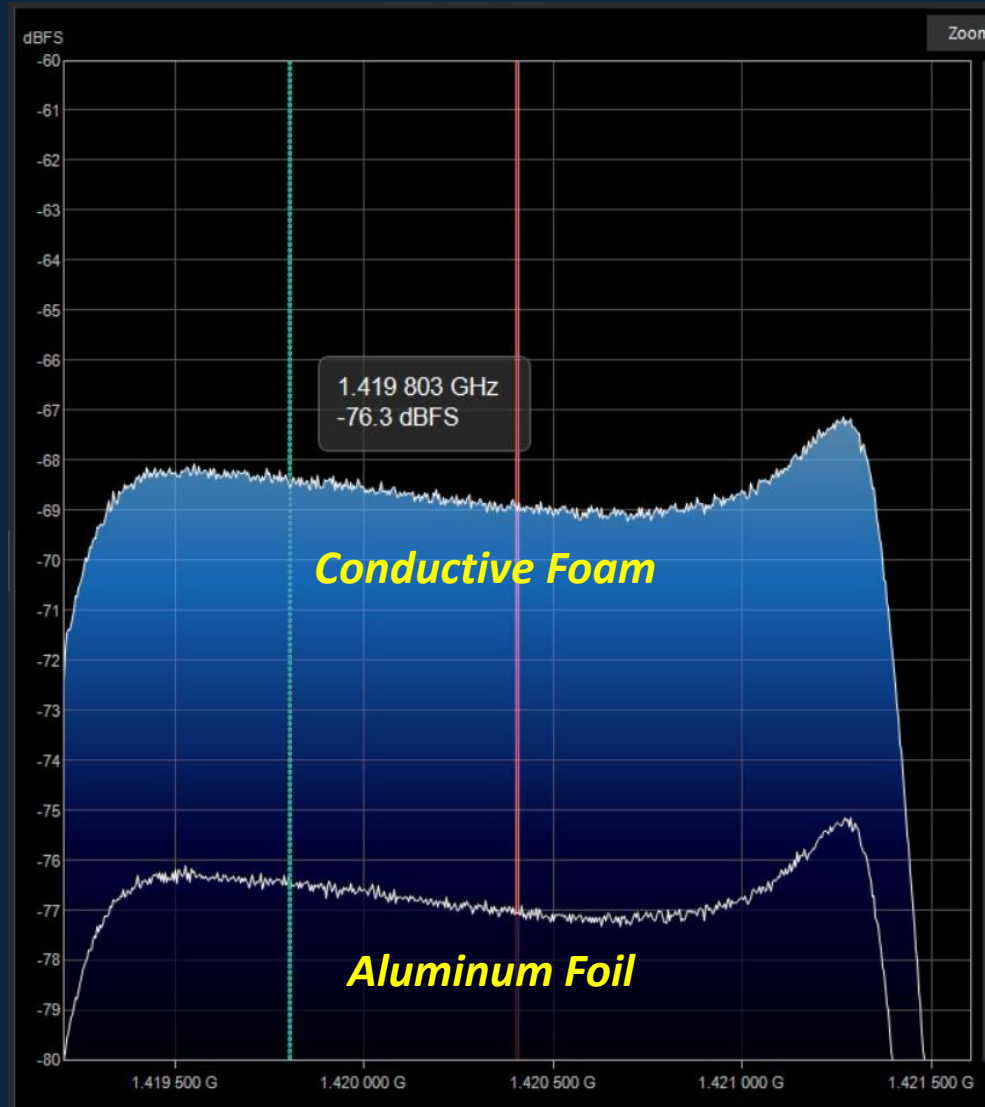
Its a perfect fit between the Flange Adapter & Feed Support Tube



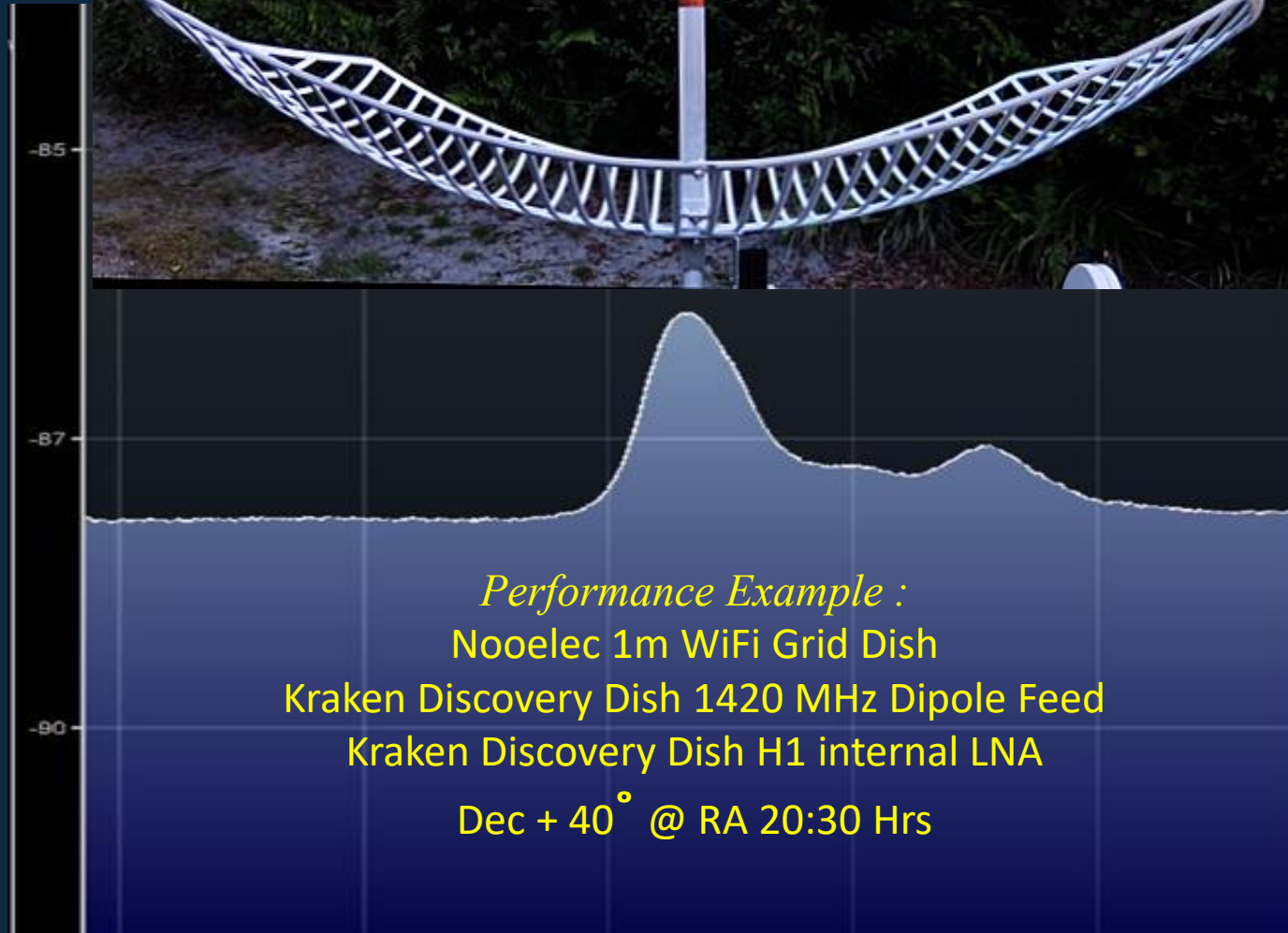
Kraken Discovery Dish 1420 MHz Feed Evaluation

IF_avg Background Correction

Materials Evaluation : Aluminum Foil works well for this application (as does a Conductive Foam Box)

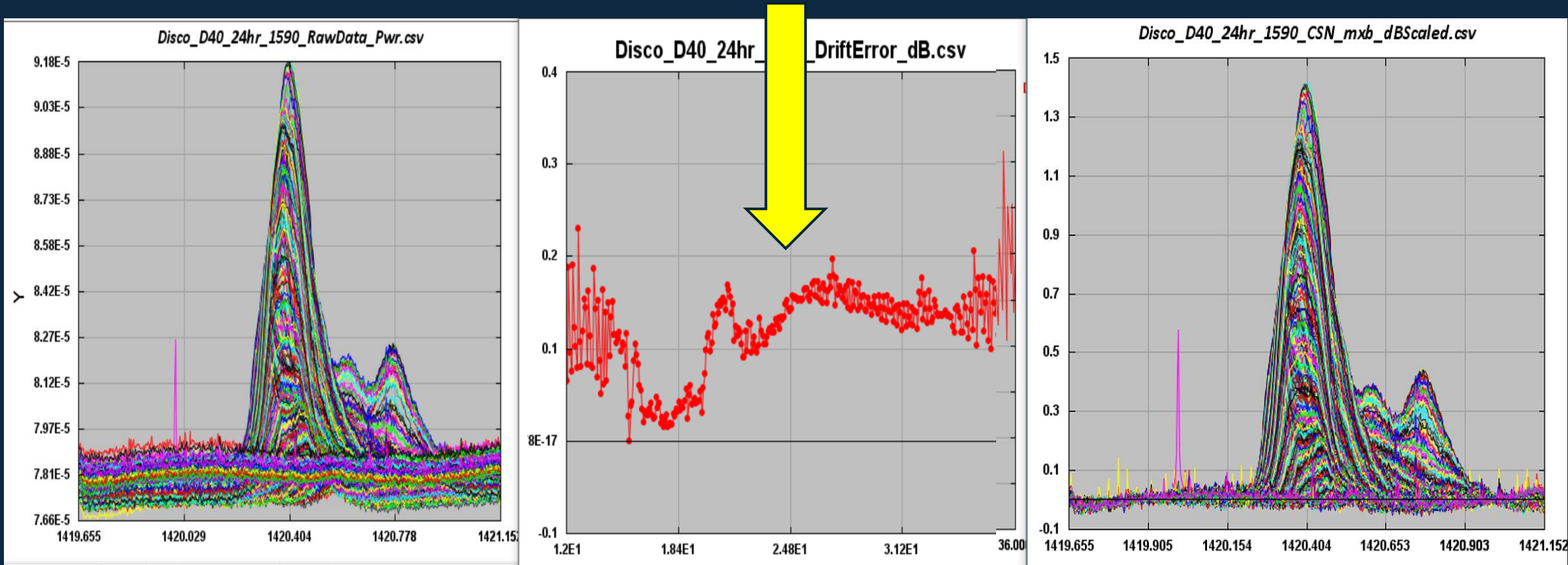


Performance Example



Noelec 1m WiFi Grid Dish
Kraken Discovery Dish 1420 MHz Dipole Feed
Kraken Discovery Dish H1 internal LNA
Dec + 40° @ 24 hour Drift Scan

Good Combination : High Quality Data & Low Frame-Frame Drift Error
(this data set has the lowest drift error I've ever measured from a 24 hour scan
with a small dish H Line Radio Telescope)

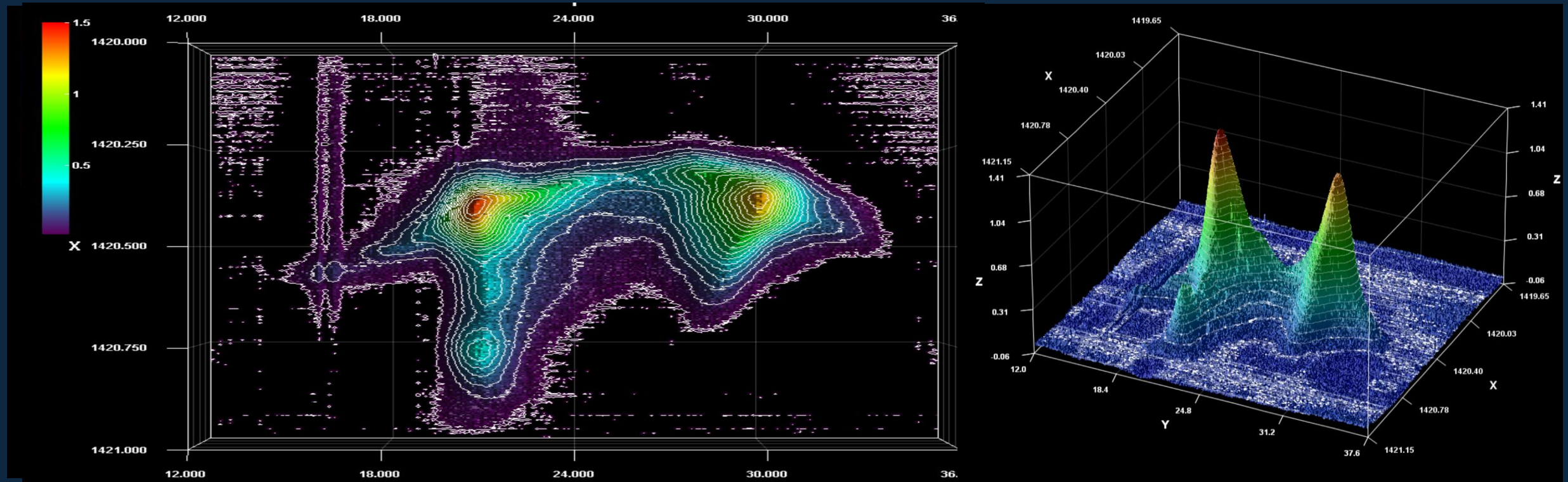


Nooelec 1m WiFi Grid Dish
Kraken Discovery Dish 1420 MHz Dipole Feed
Kraken Discovery Dish H1 internal LNA
Dec + 40° 24 hour Drift Scan

Excellent **HIGH SIGNAL** / LOW NOISE Performance

Radio Telescope Performance Evaluated Using :

AirSpy SDR# Studio > *Kaminski IF_average* > *Pettit HLine3D* > *Rinearn Graphics*
<https://github.com/AP-HLine-3D/HLine3D>

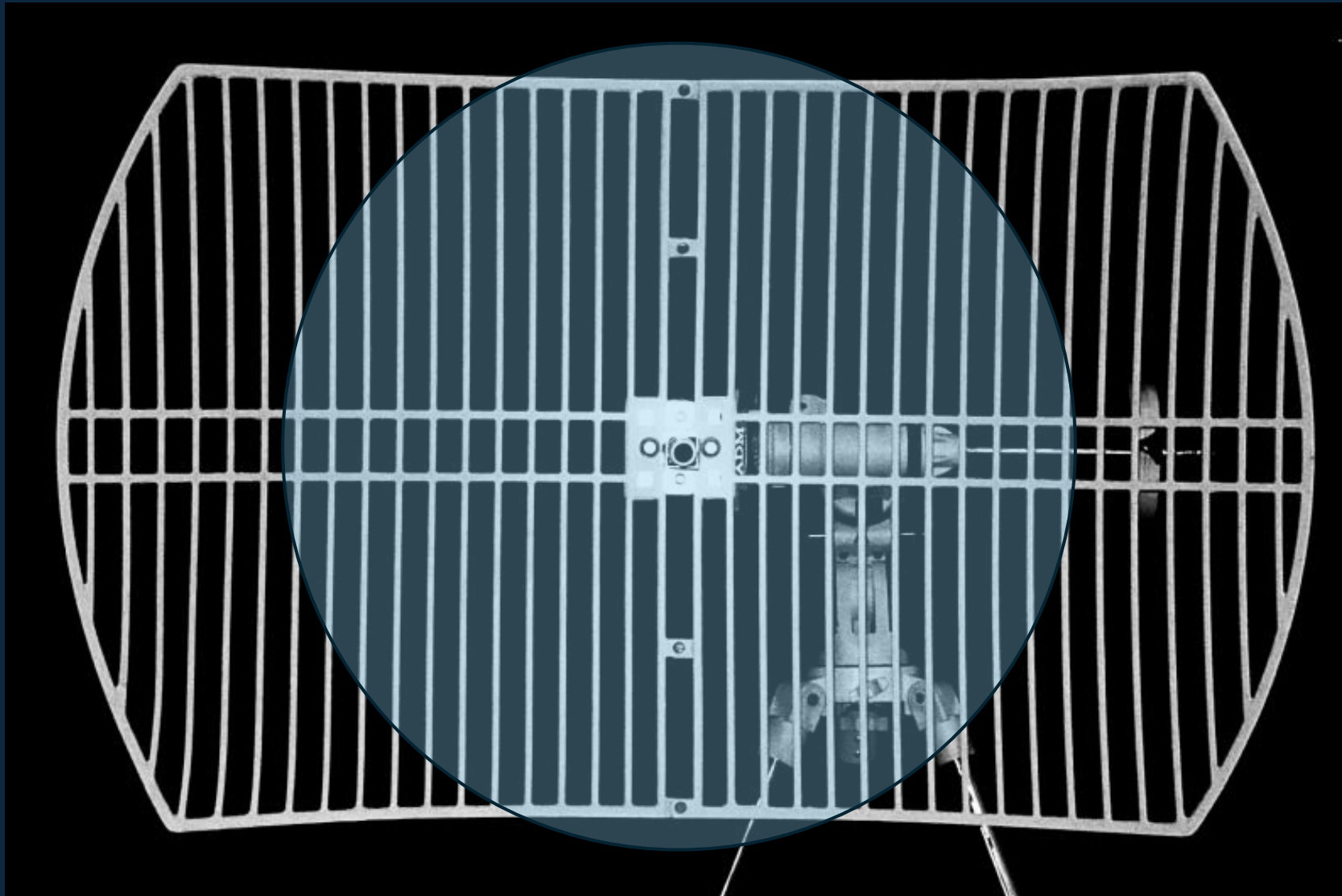


B Alex Pettit, Jr Dec 2025

Kraken Discovery Feed 1420 MHz Dish Options

1m x 0.6m WiFi Dish used for This evaluation

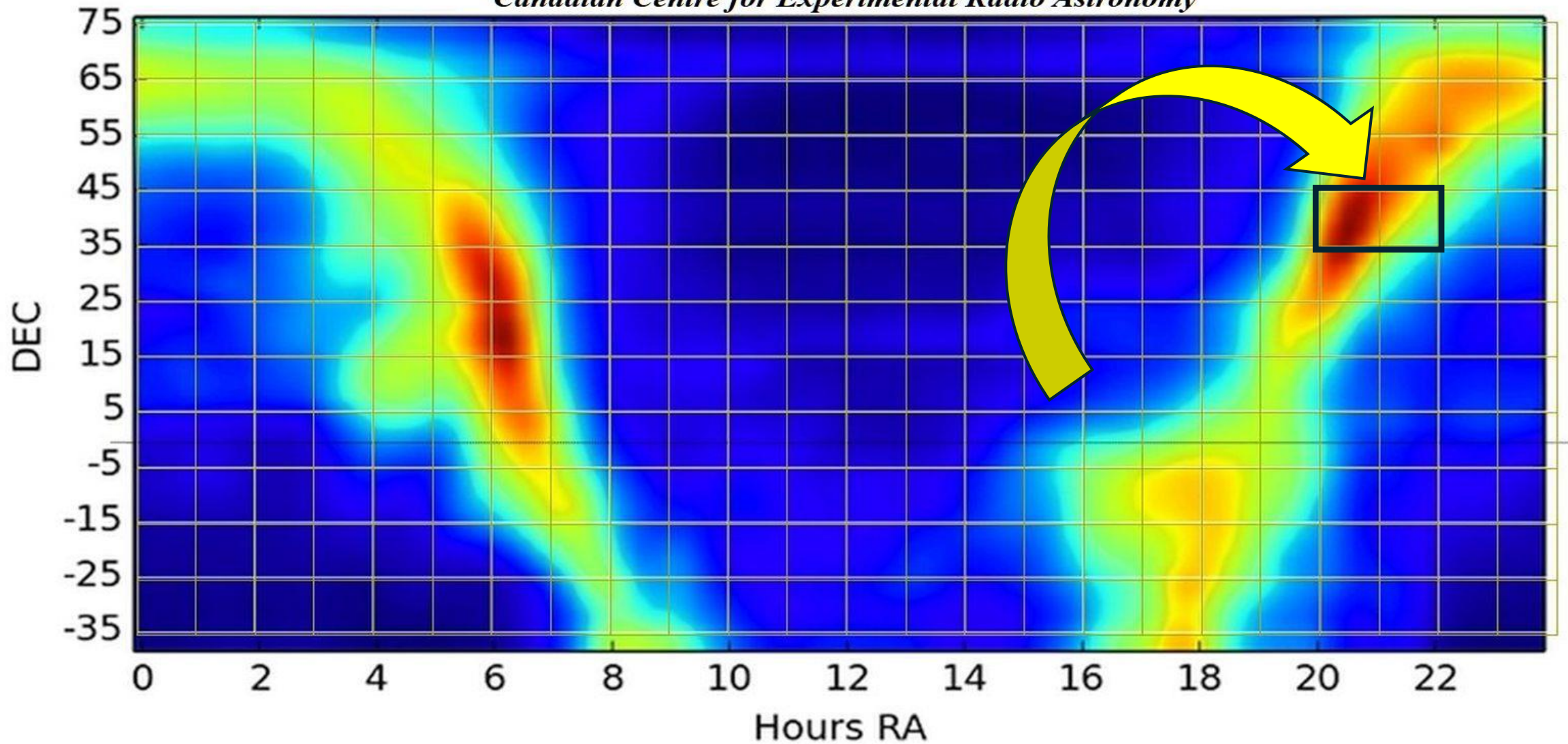
0.65m Kraken Discovery Dish will be eval'd in the future



Drift Scan 2 Hr Scan @ Declination + 40° RA 20 - 22 Hrs

CCERA

Canadian Centre for Experimental Radio Astronomy



Drift Scan 2 Hr Scan @ Declination +40° RA 20 - 22 Hrs

Disco_D40_2hrs_CSN_mxb_dBScaled.csv

